Answer **all** the questions in the spaces provided.

1 Determination of the concentration of sodium thiosulfate in solution Z

Sodium thiosulfate is commonly used in an alternative method for the extraction of gold from its ores, through the formation of a stable and soluble complex of gold(I) ions, $[Au(S_2O_3)_2]^{3-}$. One of your tasks is to determine the concentration of sodium thiosulfate in a commercially produced solution **Z** used in the extraction process.

Solution **Z** contains sodium thiosulfate in deionised water. You are not provided with solution **Z**, as it is too concentrated for use in this experiment. A diluted solution of **Z** has been prepared for you. This diluted solution is **FA 5**.

FA 1 is 0.0230 mol dm^{-3} of aqueous potassium iodate(V), KIO_3
FA 2 is 0.50 mol dm^{-3} potassium iodide, KI
FA 3 is 0.75 mol dm^{-3} sulfuric acid, H_2SO_4
FA 4 is a solution of sodium thiosulfate, $Na_2S_2O_3$ of an unknown concentration
FA 5 a diluted solution of **Z**, in which 31.00 cm^3 of solution **Z** was made up to 250 cm^3 with deionised water in a volumetric flask

You are also provided with starch solution.

In acidic medium, iodate(V) ions oxidise iodide ions to iodine

$$IO_3^-(aq) + 5I^-(aq) + 6H^+(aq) \rightarrow 3I_2(aq) + 3H_2O(l) \qquad \textbf{reaction 1}$$

If an excess of sodium thiosulfate is present in the reaction mixture, the iodine produced in **reaction 1** is immediately reduced back to iodide ions, as shown in **reaction 2**, and the dark brown colour of iodine does not form.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq) \qquad \textbf{reaction 2}$$

In the experiments that you will perform, insufficient volumes of the first sodium thiosulfate solution will be added initially, and some iodine will remain in the solution. The amount of iodine left-over can be determined by titration with the second solution of sodium thiosulfate, as shown in **reaction 2**.

You will prepare a number of reaction mixtures, each of which will contain:

- a fixed volume of **FA 1**,
- excess amounts of **FA 2** and **FA 3**,
- a different volume of **FA 4**.

All reaction mixtures will contain some iodine, produced in **reaction 1**, that will be titrated against the **FA 5** solution.

You are required to use a graphical method to determine the concentrations of sodium thiosulfate in **FA 4**, **FA 5** and solution **Z**.

(a) Titration of iodine produced against FA 5

Note: You will perform each titration only **once**. Great care must be taken to ensure that you do not overshoot the end-point.

You will be using two burettes in this task, and each of the burette will contain different sodium thiosulfate solutions, so care must be taken that the right one is used each time.

(i) Experiment 1

1. Fill the burette labelled "**FA 4**" with **FA 4** solution.
2. Fill the burette labelled "**FA 5**" with **FA 5** solution.
3. Using a pipette, transfer 25.0 cm^3 of **FA 1** into a 250 cm^3 conical flask.
   From the burette, transfer 9.50 cm^3 of **FA 4** to the same conical flask.

   If you accidentally added an incorrect volume of **FA 4** for any of your titrations, record the actual volume of **FA 4** added in your table on the next page, and use that volume for the plotting of the graph later

4. Use appropriate measuring cylinders to add to this flask:
   - 10 cm^3 of **FA 2**,
   - 10 cm^3 of **FA 3**.
5. Titrate the iodine liberated with **FA 5** until the solution fades to a pale yellow colour.
6. Add about 1 cm^3 (about 10 drops) of starch indicator to this flask. Continue adding **FA 5** until the blue-black colour **just** disappears.
7. Record your titration results in the section "Recording" on page 3. Make sure that your recorded results show the precision of your working.

(ii) Experiment 2

8. Repeat steps 3 to 8 but add 22.00 cm^3 of **FA 4** at step 4.

(iii) Experiment 3 to 5

9. Choose three further volumes of **FA 4** between 9.50 cm^3 and 22.00 cm^3.
10. Repeat steps 3 to 8 using each of your chosen volumes of **FA 4** at step 4.

Recording

Prepare a table in the space provided on the next page to record:

- the volumes of **FA 4** added for each experiment.
- the burette readings and the volumes of **FA 5** used for each experiment.

Table of results

Titration number	1	3	4	5	2
Volume of **FA 4** /cm^3	**9.50**	**12.50**	**15.50**	**18.50**	**22.00**
Final burette reading /cm^3	31.30	28.75	24.10	44.90	46.80
Initial burette reading /cm^3	0.10	1.00	0.00	24.10	30.00
Volume of **FA 5** added /cm^3	31.20	27.75	24.10	20.80	16.80

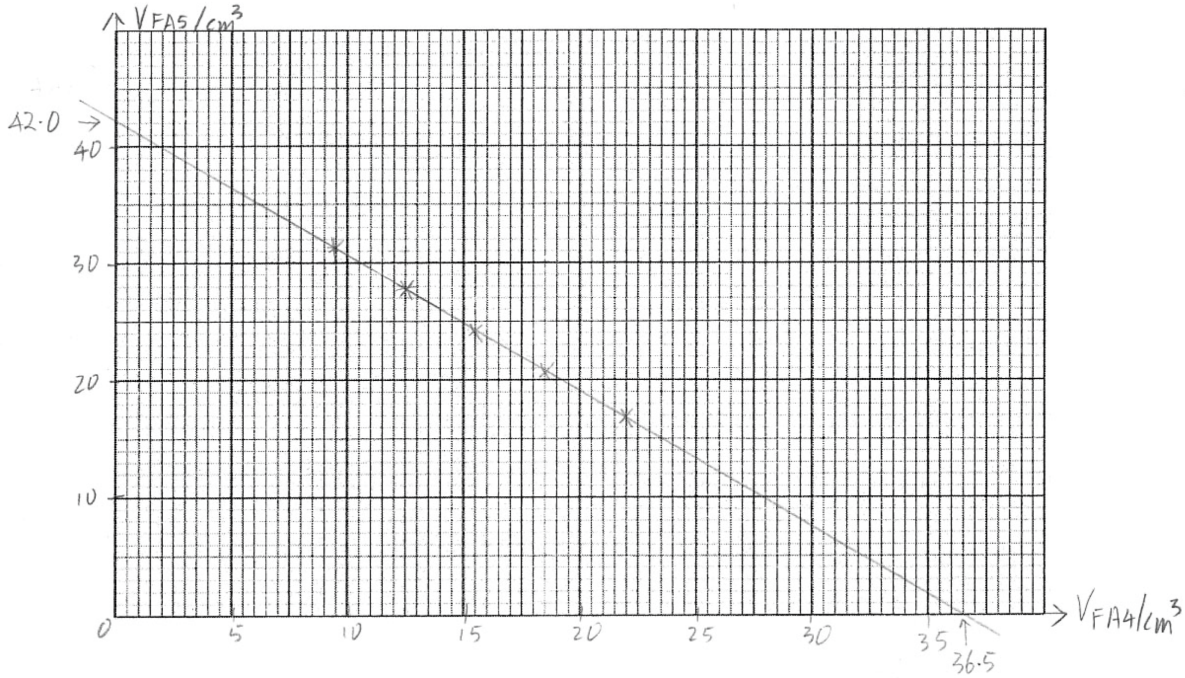
[4]

(b) Graphical determination to determine concentration of sodium thiosulfate in FA 4, FA 5 and solution Z.

(i) Plot a graph of the volume of **FA 5** added (*y-axis*) against the volume of **FA 4** added (*x-axis*) on the grid below. When choosing the scale, you should ensure that the graph line can be extrapolated to intersect both the *x-axis* and the *y-axis*.

Draw a straight line of best fit, taking into account all of your plotted points.

Extrapolate your graph line so that it intersects both the *x-axis* and the *y-axis*.



[3]

(ii) It is possible to react all the iodine produced by the reaction between 25.0 cm^3 of **FA 1** and 10 cm^3 of **FA 2** and **FA 3** by using **FA 4** only.

From your graph, determine the volume of **FA 4** required to do this.

volume of **FA 4** required = 36.5 cm^3 [3]

(iii) Calculate the gradient of your graph line, showing clearly how you did this.

gradient = $\frac{0-42.0}{36.5-0} = -1.1507 = -1.15$

gradient = −1.15 [1]

(c) Calculations

(i) The amount of iodate(V) ions, IO_3^-, present in 25.0 cm^3 of **FA 1** is 5.75×10^{-4} mol.

Use this value, together with data obtained from your graph, to calculate the concentration of thiosulfate ions in **FA 4**.

Number of moles of iodine produced = $3 \times n_{IO_3^-} = 3 \times 5.75 \times 10^{-4} = 1.725 \times 10^{-3}$ mol

Number of moles of $S_2O_3^{2-}$ required = $2 \times n_{I_2} = 2 \times 1.725 \times 10^{-3} = 3.45 \times 10^{-3}$ mol

concentration of $S_2O_3^{2-}$ in **FA4** = $\frac{n_{S_2O_3^{2-}}}{V_{\mathbf{FA4}}} = \frac{3.45 \times 10^{-3}}{36.5 \times 10^{-3}} = 0.094521 = 0.0945 \text{ mol dm}^{-3}$

$[S_2O_3^{2-}]$ in **FA 4** = 0.0945 mol dm^{-3} [2]

(ii) The concentration of $S_2O_3^{2-}$ in **FA 4** and **FA 5** are related to the gradient of the graph obtained in **(b)(iii)** by the following equation:

$$[S_2O_3^{2-}]_{\mathbf{FA\,5}} = \frac{[S_2O_3^{2-}]_{\mathbf{FA\,4}}}{|\text{gradient}|}$$

Using the equation, or otherwise, calculate the concentration of thiosulfate ions in **FA 5**.

$$[S_2O_3^{2-}]_{\mathbf{FA5}} = \frac{[S_2O_3^{2-}]_{\mathbf{FA4}}}{|\text{gradient}|} = \frac{0.094521}{1.1507} = 0.082142 = 0.0821 \text{ mol dm}^{-3}$$

$[S_2O_3^{2-}]$ in **FA 5** = 0.0821 mol dm^{-3} [1]

(iii) Calculate the concentration of thiosulfate ions in the commercially produced solution **Z**.

$$[S_2O_3^{\ 2-}]_z \times 31.00\times10^{-3} = [S_2O_3^{\ 2-}]_{\mathbf{FA5}} \times 250\times10^{-3}$$

$$\Rightarrow [S_2O_3^{\ 2-}]_{\mathbf{z}} = \frac{[S_2O_3^{\ 2-}]_{\mathbf{FA5}} \times 250\times10^{-3}}{31.00\times10^{-3}} = \frac{0.082142\times250\times10^{-3}}{31.00\times10^{-3}} = 0.662 \text{ mol dm}^{-3}$$

$[S_2O_3^{2-}]$ in solution **Z** = 0.622 mol dm^{-3} [1]

(d) A student suggested using burettes, rather than using measuring cylinders, to measure the volume of **FA 2** and **FA 3** used in **(a)** so as to improve on the accuracy of the titration data.

State whether you agree with the student and briefly explain why.

No. Since both solutions are in excess, using a burette (which has a higher precision than measuring cylinders) does not improve the accuracy of the experiment.

[1]

(e) In a private institution, a laboratory assistant made the mistake in the preparation of the **FA 5** solution by diluting 21.00 cm^3 of solution **Z** to 250 cm^3, rather than the specified volume of 31.00 cm^3.

State and explain how using this incorrect sample of **FA 5** will affect the gradient of your graph.

As the concentration of $S_2O_3^{2-}$ in the incorrect sample of **FA 5** will be **lower** than the intended value, and since $[S_2O_3^{\ 2-}]_{\mathbf{FA\,5}} = \dfrac{[S_2O_3^{\ 2-}]_{\mathbf{FA\,4}}}{|\text{gradient}|}$, the **magnitude of the gradient** will be **larger**.

Or

As the concentration of $S_2O_3^{2-}$ in the incorrect sample of **FA 5** will be lower than the intended value, a larger volume of **FA 5** would be required for each titration, leading to a larger magnitude for the gradient.

[1]

[Total: 17]

2 Determination of the enthalpy change for the thermal decomposition of sodium hydrogen carbonate

FA 6 is solid anhydrous sodium carbonate, Na_2CO_3

FA 7 is solid sodium hydrogen carbonate, $NaHCO_3$

You will need access to the **FA 3** solution that you have used earlier.

Sodium hydrogen carbonate is commonly used as a raising agent in baking, as it undergoes thermal decomposition at temperatures above 80 °C, according to the equation:

$$2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l) \qquad \textbf{reaction 1}$$

Due to the high temperature involved, the enthalpy change of this reaction cannot be measured directly using a coffee-cup calorimeter.

In the experiments that you will perform, the coffee-cup calorimeter will be used to determine the enthalpy change, ΔH_2, for the reaction:

$$Na_2CO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + CO_2(g) + H_2O(l) \qquad \textbf{reaction 2}$$

and the enthalpy change, ΔH_3, for the reaction:

$$2NaHCO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2CO_2(g) + 2H_2O(l) \qquad \textbf{reaction 3}$$

You will then use your results of your experiments to calculate the enthalpy change, ΔH_1, for the thermal decomposition of sodium hydrogen carbonate.

(a) Determining enthalpy change for the reaction between Na_2CO_3 and H_2SO_4, ΔH_2

1. Place the styrofoam cup inside a 250 cm^3 beaker to prevent it from toppling over.
2. Using a measuring cylinder, place 40 cm^3 of **FA 3** into the styrofoam cup.
3. Stir the solution in the cup gently with the thermometer, and measure its temperature. Record the initial temperature of **FA 3** in your table.
4. Weigh the weighing bottle labelled "**FA 6**" and its contents using a weighing scale, without the cap. The bottle has been pre-weighed to contain between 4.50 g and 5.00 g of solid sodium carbonate.
5. Pour **FA 6** from the weighing bottle into the styrofoam cup, place the lid onto the cup, and insert the thermometer through the lid.
6. Stir the mixture using the thermometer and record the maximum temperature reached.
7. Reweigh the weighing bottle.

(i) In an appropriate format in the space below, record:

- all mass readings, including the mass of **FA 6** added to the styrofoam cup
- all temperatures measured and the temperature rise.

Mass of weighing bottle and **FA 6**	/ g	8.104
Mass of weighing bottle and residual **FA 6**	/ g	3.542
Mass of **FA 6** added to the styrofoam cup	/ g	4.562
Initial temperature of **FA 3**	/ °C	30.2
Maximum temperature reached	/ °C	39.2
Temperature rise	/ °C	9.0

[1]

(ii) Calculate the heat change for **reaction 2**.

Assume that the specific heat capacity of the final solution is 3.75 J g^{-1} K^{-1} and that its density is 1.00 g cm^{-3}.

$$q = (40.0 \times 1.00) \times 3.75 \times 9.0 = 1350 \text{ J} = 1.35 \text{ kJ}$$

heat change = 1.35 kJ [1]

(iii) By calculating the number of moles of each reagent that was added to the styrofoam cup, and with reference to the equation for the reaction, show which of the reagents, **FA 3** or **FA 6**, is added in excess.

[A_r: Na, 23.0; C, 12.0; O, 16.0]

$$n_{H_2SO_4} = 0.75 \times 40.0 \times 10^{-3} = 0.0300 \text{ mol}$$

$$n_{Na_2CO_3} = \frac{4.562}{2(23.0) + 12.0 + 3(16.0)} = \frac{4.562}{106.0} = 0.043038 \text{ mol}$$

From the balanced equation, $\dfrac{n_{H_2SO_4}}{n_{Na_2CO_3}} = \dfrac{1}{1}$,

0.0300 mol of H_2SO_4 will require only 0.0300 mol of Na_2CO_3 for complete reaction, and since the number of moles of Na_2CO_3 added is more than 0.0300 mol, Na_2CO_3 is added in excess.

reagent added in excess = Na_2CO_3 [1]

(iv) Calculate enthalpy change for **reaction 2**, ΔH_2:

$$Na_2CO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + CO_2(g) + H_2O(l)$$

$$\Delta H_{rxn} = -\frac{1.35}{0.0300} = -45.0 \text{ kJ mol}^{-1}$$

ΔH_2 = −45.0 kJ mol^{-1} [1]

(b) Determining enthalpy change for the reaction between $NaHCO_3$ and H_2SO_4, ΔH_2

1. Empty and rinse the styrofoam cup used in **(a)**, and replace the cup in the 250 cm^3 beaker.
2. Using the same measuring cylinder used in **(a)**, place 40 cm^3 of **FA 3** into the styrofoam cup.
3. Wash and dry the thermometer used in **(a)**, before using it to stir the solution in the cup gently, and measure its temperature. Record the initial temperature of **FA 3** in your table.
4. Weigh the weighing bottle labelled "**FA 7**" and its contents using a weighing scale, without the cap. The bottle has been pre-weighed to contain between 3.50 g and 4.00 g of solid sodium hydrogen carbonate.
5. Pour **FA 7** from the weighing bottle into the styrofoam cup, place the lid onto the cup, and insert the thermometer through the lid.
6. Stir the mixture using the thermometer and record the minimum temperature reached.
7. Reweigh the weighing bottle.

(i) In an appropriate format in the space below, record:

- all mass readings, including the mass of **FA 7** added to the styrofoam cup
- all temperatures measured and the temperature fall.

Mass of weighing bottle and **FA 7**	/ g	7.428
Mass of weighing bottle and residual **FA 7**	/ g	3.463
Mass of **FA 7** added to the styrofoam cup	/ g	3.965
Initial temperature of **FA 3**	/ °C	30.2
Minimum temperature reached	/ °C	37.8
Temperature fall	/ °C	7.6

[3]

(ii) Calculate the heat change for **reaction 3**.

Assume that the specific heat capacity of the final solution is 3.75 J g^{-1} K^{-1} and that its density is 1.00 g cm^{-3}.

$$q = (40.0 \times 1.00) \times 3.75 \times 7.6 = 1140 \text{ J} = 1.14 \text{ kJ}$$

heat change = 1.14 kJ [1]

(iii) By calculating the number of moles of each reagent that was added to the styrofoam cup, and with reference to the equation for the reaction, show which of the reagents, **FA 3** or **FA 7**, is added in excess.

[A_r: Na, 23.0; H, 1.0; C, 12.0; O, 16.0]

$$n_{H_2SO_4} = 0.75 \times 40.0 \times 10^{-3} = 0.0300 \text{ mol}$$

$$n_{NaHCO_3} = \frac{3.965}{23.0 + 1.0 + 12.0 + 3(16.0)} = \frac{3.965}{84.0} = 0.047202 \text{ mol}$$

From the balanced equation, $\frac{n_{H_2SO_4}}{n_{NaHCO_3}} = \frac{1}{2}$,

0.0300 mol of H_2SO_4 will require 0.0600 mol of $NaHCO_3$ for complete reaction, and since the number of moles of $NaHCO_3$ added is less than 0.0600 mol, H_2SO_4 is added in excess.

reagent added in excess = H_2SO_4 [1]

(iv) Calculate enthalpy change for **reaction 3**, ΔH_3:

$$2NaHCO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2CO_2(g) + 2H_2O(l)$$

$$\Delta H_{rxn} = +\frac{1.14}{0.047202} \times 2 = +48.303 = +48.3 \text{ kJ mol}^{-1}$$

ΔH_3 = $+48.3$ kJ mol^{-1} [1]

(c) Use your answer to **(a)(iv)** and **(b)(iv)** to calculate enthalpy change for **reaction 1**, ΔH_1:

$$2NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$$

$$2NaHCO_3(s) \xrightarrow{\Delta H_1} Na_2CO_3(s) + CO_2(g) + H_2O(l)$$

ΔH_3 + $H_2SO_4(aq)$ $\qquad$ ΔH_2 + $H_2SO_4(aq)$

$$Na_2SO_4(aq) + 2CO_2(g) + 2H_2O(l)$$

Hence, $\Delta H_1 = \Delta H_3 - \Delta H_2 = +48.303 - (-45.0) = +93.303 = +93.3 \text{ kJ mol}^{-1}$

ΔH_1 = $+93.3$ kJ mol^{-1} [4]

(d) In **(a)**, one of the significant sources of errors that will affect the accuracy of the enthalpy change for the reaction between Na_2CO_3 and H_2SO_4, ΔH_2, is that the heat capacity of the styrofoam cup is not taken into account.

Explain the effect of this error on your calculated value of ΔH_2.

As some of the **heat produced** in the reaction **is absorbed** by the styrofoam cup, the **maximum temperature** measured is **lower** than expected, and so the **magnitude** of ΔH_2 calculated will be **lower** than the theoretical value

[1]

[Total: 15]

3 Planning

A sample of solid anhydrous sodium carbonate is believed to be contaminated with 5 to 10% of sodium chloride, Na*Cl*.

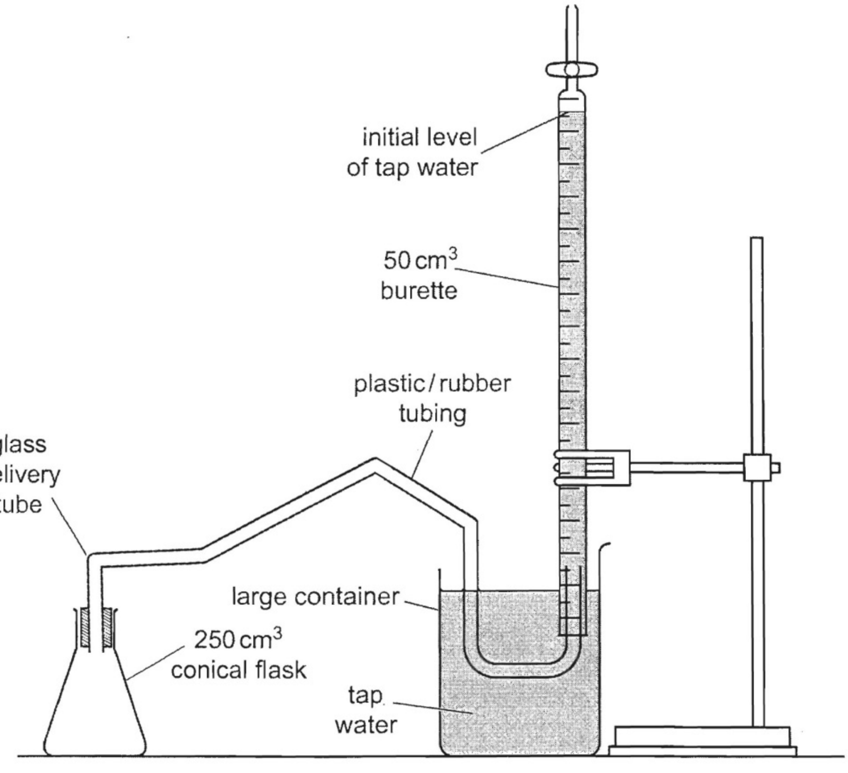
The percentage purity of sodium carbonate in the sample can be determined by measuring the volume of carbon dioxide formed when sodium carbonate is reacted with an excess of sulfuric acid.

$$Na_2CO_3(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + CO_2(g) + H_2O(l)$$

You are to plan an experiment that will enable you to:

- react a pre-weighed impure sample of sodium carbonate with a pre-determined volume of the **FA 3** solution that you have used earlier for Question 1 and 2,
- collect and measure the volume of the carbon dioxide gas evolved.

(a) Draw a diagram of the apparatus that you would use to carry out this experiment.



[2]

(b) By considering the capacity of the apparatus that you have drawn in **(a)** to measure the volume of the carbon dioxide gas, calculate the maximum mass of the impure sample of sodium carbonate that you would use in your experiment.

[A_r: Na, 23.0; C, 12.0; O, 16.0 and 1 mole of gas occupies a volume of 24.0 dm^3 under the conditions present in the laboratory]

Taking the maximum volume of CO_2(g) collected to be 50 cm^3,

$$n_{CO_2} = \frac{50}{24000} = 2.0833 \times 10^{-3} \text{ mol}$$

$$\Rightarrow n_{Na_2CO_3} = 2.0833 \times 10^{-3} \text{ mol}$$

$$m_{Na_2CO_3} = 2.0833 \times 10^{-3} \times \{2(23.0) + 12.0 + 3(16.0)\} = 2.0833 \times 10^{-3} \times \{106.0\} = 0.22083 = 0.221 \text{ g}$$

maximum mass of impure sample of sodium carbonate used = 0.221 g [2]

(c) A student suggested that 40.0 cm^3 of **FA 3** solution is sufficient for the experiment.

State whether you agree with the student, and explain your answer.

Yes, I agree with the student. The number of moles of H_2SO_4 present in 40.0 cm^3 of **FA 3** {0.0300 mol, calculated in **2(a)(iii)** and **2(b)(iii)**} is more than enough to react with the amount of Na_2CO_3 calculated in **3(b)**.

[1]

(d) Describe briefly how you would ensure that all the carbon dioxide gas given off in the reaction between the impure sample of sodium carbonate and **FA 3** solution was collected and that none escaped from the apparatus.

Place the weighed sample of sodium carbonate in the conical flask, and stopper the flask. Introduce the measured volume of **FA 3** solution using a thistle funnel (the volume of **FA 3** introduced need to be subtracted from the volume of gas collected).

or

Place the weighing bottle that contains the weighed sample of sodium carbonate (carefully) into the conical flask that contains the measured volume of **FA 3** solution. Stopper the conical flask. Tilt the conical flask such that the weighing bottle is tipped over in order to start the reaction.

[1]

(e) Other than this method that involves the collection of carbon dioxide gas, the percentage purity of sodium carbonate in the sample can also be determined using a titration.

In a typical experiment, a pre-weighed mass of the impure sample of sodium carbonate is dissolved in water and made up to 250 cm^3 in a volumetric flask, and 25.0 cm^3 of this solution is titrated against a standard solution of hydrochloric acid using methyl orange as indicator.

State which of the two methods you would expect to give the more accurate results. Explain your choice.

Titration method.

Source of error for the collection of CO_2 gas method:

- The volume of the CO_2 gas may not be measured under standard conditions (the volume collected will be lower than the actual volume produced, as the gas syringe may not be perfectly frictionless; or the volume measured will be higher than the actual volume as the water level inside the burette is higher than the water level in the beaker for the downward displacement of water).

Other sources of errors if downward displacement of water method was used:

- CO_2 gas is slightly soluble in water (the volume collected will be lower than the actual volume produced, as some of the CO_2 produced may be dissolved in the water used for downward displacement of water).
- The rubber tubing contains a certain volume of water at the start of the experiment that must first be displaced before the CO_2 gas can be collected (the volume collected will be lower than the actual volume produced).

[2]

[Total: 8]

4 Inorganic Qualitative Analysis

You are provided with **FA 8**, which is a mixture of two solids, **FA 9**, which is soluble in water and **FA 10**, which is insoluble in water. Each contains **one cation** and **one anion** from the lists on page 16 and 17.

In this question you will perform a series of test-tube reactions, make observations and deduce the identities of the cations and anions present in **FA 9** and **FA 10**.

If the evolution of a gas is observed at any stage, the gas should be tested and identified.

You are advised that the reagent should be added gradually in all tests, with shaking after each addition.

If it appears that no reaction has taken place, this should be clearly recorded.

(a) Tests on FA 8

Perform the test-tube experiments described below and record your observations in the spaces provided in the table.

	Test	Observations
(i)	Place all of the solid, **FA 8**, into a 100 cm^3 beaker. Add 25 cm^3 of distilled water and warm to dissolve the **FA 9**. Filter the mixture and use the filtrate for tests **(a)(ii)** to **(a)(iv)**. Wash the residue, **FA 10**, and use it for tests **(a)(v)** to **(a)(vii)**.	A green residue **[✓1]** and a colourless filtrate was obtained **[✓2]**.

Tests on Filtrate, FA 9

	Test	Observations
(ii)	Place about 1 cm depth of **FA 9** into a test-tube. Add a few drops of dilute sulfuric acid.	A white ppt is formed **[✓3]**.
(iii)	Place about 1 cm depth of **FA 9** into a test-tube. Add a few drops of aqueous barium chloride, followed by a few drops of dilute hydrochloric acid.	No ppt is formed **[✓4]**. No ppt is formed / no gas is produced **[✓5]**.
(iv)	Place about 1 cm depth of **FA 9** into a test-tube. Add a few drops of aqueous silver nitrate. followed by a few drops of aqueous ammonia.	A white ppt is formed **[✓6]**. The ppt dissolves to give a colourless solution **[✓7]**.

Tests on Residue, FA 10

	Test	**Observations**
(v)	Using a spatula, transfer as much of the residue on the filter paper as you can into a test-tube. Add dilute sulfuric acid drop-wise into the test-tube to dissolve the residue. Filter if necessary. Use the resultant solution for test **(a)(vi)** and **(a)(vii)**.	Effervescence produced **[✓8]**, the gas gives white ppt with limewater **[✓9]**. A pale green / blue solution is produced **[✓10]**.
(vi)	Place about 1 cm depth of the **filtrate** from **(a)(v)** in a test-tube. Add aqueous sodium hydroxide drop-wise with shaking, until no further change is seen.	A (pale) blue ppt is formed that is insoluble in excess NaOH(aq) **[✓11]**.
(vii)	Place about 1 cm depth of the **filtrate** from **(a)(v)** into a test-tube. Add aqueous ammonia, drop-wise with shaking, until no further change is seen.	A blue ppt is formed **[✓12]** that is soluble in excess NH_3(aq) to form a dark blue solution **[✓13]**.

[5]

(b) Based on your observations to **(a)(ii)**, suggest a possible identity of the cation in **FA 9**.

cation in **FA 9**: Ba^{2+} [1]

(c) From your remaining observations, state the identities of the one anion in both **FA 9** and **FA 10**, and the one cation in **FA 10**.

In each case, give evidence to support your conclusion.

	identity		evidence
FA 9	anion:	Cl^-	A white ppt is formed with $AgNO_3$(aq), that is soluble in NH_3(aq) in **(a)(iv)**.
FA 10	cation:	Cu^{2+}	A (pale) blue ppt is formed with NaOH(aq) that is insoluble in excess in **(a)(vi)**. **<u>Or</u>** A blue ppt is formed with NH_3(aq) that is soluble in excess to form a dark blue solution in **(a)(vii)**.
	anion:	CO_3^{2-}	CO_2(g) produced with H_2SO_4(aq) in **(a)(v)**.

[3]

[Total: 9]

5 Planning

In this question, you will devise a plan, using test-tube reactions, to distinguish between four organic compounds, so that each is identified.

Consider 4 unlabelled bottles and each bottle contains one of the following colourless liquids:

ethanal propanone propan-2-ol ethanoic acid

Plan an investigation, using test tube reactions, which would allow you to identify each of these organic compounds.

Each compound should be identified by at least one positive test. It is not sufficient to identify a compound simply by eliminating all the others.

Your plan should include:

- details of the reagents and conditions to be used,
- an outline of the sequence of steps you would follow,
- an explanation of how you would analyse your results in order to identify each compound.

Once a compound has been clearly identified, your plan should concentrate on distinguishing the remaining compounds.

	Test	**Expected Observations**
1	Place about 1 cm depth of each of the four samples into four separate test tubes. Add a few drops aqueous sodium carbonate.	If effervescence is produced, and the gas gives a white ppt with limewater, the sample is ethanoic acid. If no effervescence is produced, the sample is ethanal, propanone or propan-2-ol.
2	Place about 1 cm depth of a fresh sample for each of the three remaining samples into three separate test tubes. Add a few drops of Tollen's reagent, and warm in a hot water bath.	If a grey ppt or a silver mirror is formed, the sample is ethanal. If no grey ppt or silver mirror is obtained, the sample is propanone or propan-2-ol.
3	Place about 1 cm depth of a fresh sample for each of the two remaining samples into two separate test-tubes. Add a few drops of 2,4-dinitrophenylhydrazine.	If a yellow / orange ppt is formed, the sample is propanone If no ppt is formed, the same is probably propan-2-ol.
4	Place about 1 cm depth of a fresh sample for the last sample in a test-tube. Add 10 drops of dilute sulfuric acid, followed by 2 drops of potassium manganate(VII) and warm in the water bath.	If the purple $KMnO_4$ is decolourised, the sample is propan-2-ol.

[Total: 6]

Qualitative Analysis Notes
[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	NaOH(aq)	NH_3(aq)
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq),	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt. insoluble in excess	green ppt. insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt. insoluble in excess	off-white ppt. insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

ions	***reaction***
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^-(aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^-(aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^-(aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil
nitrite, NO_2^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil; NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, SO_4^{2-}(aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-}(aq)	SO_2 liberated with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Test for gases

ions	***reaction***
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	***colour of element***	***colour in aqueous solution***	***colour in hexane***
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

END OF PAPER 4

Preparation list

Chemicals:

FA 1 is 0.0230 mol dm^{-3} of potassium iodate(V), KIO_3 (about 150 cm^3 per student)
FA 2 is 0.500 mol dm^{-3} potassium iodide, KI (about 70 cm^3 per student)
FA 3 is 0.750 mol dm^{-3} dilute sulfuric acid (about 230 cm^3 per student)
FA 4 is 0.100 mol dm^{-3} sodium thiosulfate, $Na_2S_2O_3$ (about 120 cm^3 per student)
FA 5 is 0.0869 mol dm^{-3} sodium thiosulfate, $Na_2S_2O_3$ (about 180 cm^3 per student)

starch solution containing 20 g dm^{-3} of soluble starch (about 10 cm^3 per student)

FA 6 is solid anhydrous sodium carbonate, Na_2CO_3 (preweighed 4.50 − 5.00 g)
FA 7 is solid sodium hydrogen carbonate, $NaHCO_3$ (preweighed 3.50 − 4.00 g)

FA 8 is a 10 : 1 solid mixture of $CuCO_3 \cdot Cu(OH)_2 \cdot H_2O$ and $BaCl_2 \cdot 2H_2O$
(preweighed 2.00 − 2.50 g, mixture should be thoroughly mixed)

Apparatus:

1 × 50 cm^3 burette labelled **FA 4**
1 × 50 cm^3 burette labelled **FA 5**
1 × 25.0 cm^3 pipette with pipette filler
2 × 250 cm^3 conical flask
1 × 10 cm^3 measuring cylinder
1 × 25 cm^3 measuring cylinder
1 × 50 cm^3 measuring cylinder
1 × 100 cm^3 beaker
1 × 250 cm^3 beaker
1 × 250 cm^3 beaker labelled “waste”

2 × glass filter funnel
1 × Styrofoam cup with lid
1 × thermometer (0.2 °C graduation)
1 × distilled water bottle (with access to additional supplies of deionised water)
1 × retort stand with double clamps
1 × paper towel

6 × test-tubes
2 × small test-tubes
1 × boiling tube
6 × dropper / teat pipette
1 × glass rod and spatula
1 × set of blue / red litmus paper + filter paper strips and box of filter paper
1 × tripod stand, wire gauze, and heat-proof mat
1 × bunsen burner and lighter

weighing scale (for weighing / reweighing of **FA 6** & **FA 7**, about 1 per 8-12 candidates)

Bench Reagents:

aqueous sodium hydroxide (approximately 2.0 mol dm^{-3})
aqueous ammonia (approximately 2.0 mol dm^{-3})
hydrochloric acid (approximately 2.0 mol dm^{-3})
nitric acid (approximately 2.0 mol dm^{-3})
sulfuric acid (approximately 1.0 mol dm^{-3})
aqueous silver nitrate (approximately 0.05 mol dm^{-3})
aqueous barium chloride (approximately 0.2 mol dm^{-3})
limewater (saturated solution of calcium hydroxide)